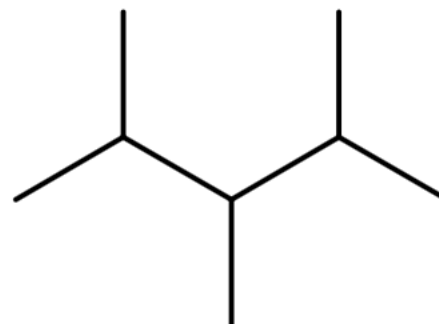


QX4001 – Molecules to Materials

Rotation 2 – Question Set 2

1) (Exercise 5.7 page 251)

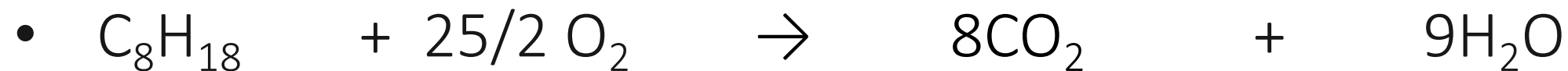
Gasoline is composed primarily of hydrocarbons, including many with eight carbon atoms, called octanes. One of the cleanest-burning octanes is a compound called 2,3,4- trimethylpentane (C_8H_{18}), which has the following structural formula:



The complete combustion of one mole of this compound to $CO_2(g)$ and $H_2O(g)$ leads to $\Delta H^\circ = -5064.9 \text{ kJ}$.

- (a) Write a balanced equation for the combustion of 1 mol of $C_8H_{18}(l)$.
- (b) By using the information in this problem and data in Table 5.3 in the book (page 231), calculate ΔH_f° for 2,3,4-trimethylpentane.

(a) Write a balanced equation for the combustion of 1 mol of $\text{C}_8\text{H}_{18}(\text{l})$.

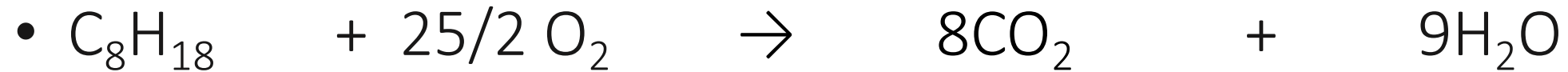


(b) By using the information in this problem and data in Table 5.3 in the book (page 231), calculate ΔH_f° for 2,3,4-trimethylpentane.

TABLE 5.3 Standard Enthalpies of Formation, ΔH_f° , at 298 K

Substance	Formula	ΔH_f° (kJ/mol)	Substance	Formula	ΔH_f° (kJ/mol)
Acetylene	$\text{C}_2\text{H}_2(g)$	226.7	Hydrogen chloride	$\text{HCl}(g)$	-92.30
Ammonia	$\text{NH}_3(g)$	-46.19	Hydrogen fluoride	$\text{HF}(g)$	-268.60
Benzene	$\text{C}_6\text{H}_6(l)$	49.0	Hydrogen iodide	$\text{HI}(g)$	25.9
Calcium carbonate	$\text{CaCO}_3(s)$	-1207.1	Methane	$\text{CH}_4(g)$	-74.80
Calcium oxide	$\text{CaO}(s)$	-635.5	Methanol	$\text{CH}_3\text{OH}(l)$	-238.6
Carbon dioxide	$\text{CO}_2(g)$	-393.5	Propane	$\text{C}_3\text{H}_8(g)$	-103.85
Carbon monoxide	$\text{CO}(g)$	-110.5	Silver chloride	$\text{AgCl}(s)$	-127.0
Diamond	$\text{C}(s)$	1.88	Sodium bicarbonate	$\text{NaHCO}_3(s)$	-947.7
Ethane	$\text{C}_2\text{H}_6(g)$	-84.68	Sodium carbonate	$\text{Na}_2\text{CO}_3(s)$	-1130.9
Ethanol	$\text{C}_2\text{H}_5\text{OH}(l)$	-277.7	Sodium chloride	$\text{NaCl}(s)$	-410.9
Ethylene	$\text{C}_2\text{H}_4(g)$	52.30	Sucrose	$\text{C}_{12}\text{H}_{22}\text{O}_{11}(s)$	-2221
Glucose	$\text{C}_6\text{H}_{12}\text{O}_6(s)$	-1273	Water	$\text{H}_2\text{O}(l)$	-285.8
Hydrogen bromide	$\text{HBr}(g)$	-36.23	Water vapor	$\text{H}_2\text{O}(g)$	-241.8

(b) By using the information in this problem and data in Table 5.3 in the book (page 231), calculate ΔH_f° for 2,3,4-trimethylpentane.



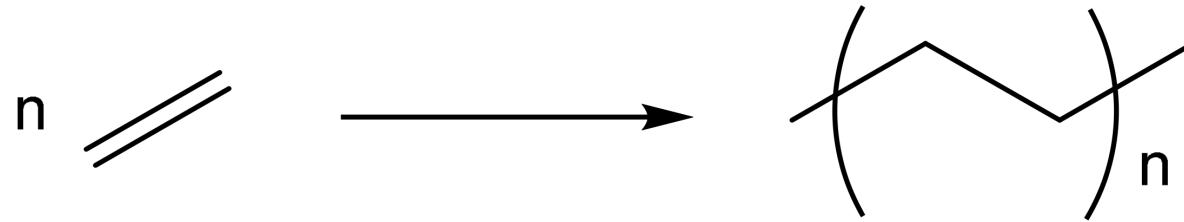
$$\bullet \Delta H^\circ_{\text{rxn}} = 8 \Delta H^\circ_{\text{CO}_2} + 9\Delta H^\circ_{\text{H}_2\text{O}} - 25/2 \Delta H^\circ_{\text{O}_2} - \Delta H^\circ_{\text{C}_8\text{H}_{18}}$$

$$\bullet \Delta H^\circ_{\text{C}_8\text{H}_{18}} = 8 \Delta H^\circ_{\text{CO}_2} + 9\Delta H^\circ_{\text{H}_2\text{O}} - 25/2 \Delta H^\circ_{\text{O}_2} - \Delta H^\circ_{\text{rxn}}$$

$$\bullet \Delta H^\circ_{\text{C}_8\text{H}_{18}} = 8 (-393.5 \text{ kJ}) + 9 (-241.8 \text{ kJ}) - 0 - (-5064.9 \text{ kJ})$$

$$\bullet \Delta H^\circ_{\text{C}_8\text{H}_{18}} = -259.3 \text{ kJ}$$

2) Consider the following polymerisation reaction:



- a) Explain why ΔH° and ΔS° are negative ($\Delta H^\circ = -109 \text{ kJ mol}^{-1}$; $\Delta S^\circ = -155 \text{ J mol}^{-1}\text{K}^{-1}$)
- b) Calculate ΔG°
- c) Calculate at which temperature polymerisation stops being spontaneous
- d) What limitations does the calculation in c) have?

a) Explain why ΔH° and ΔS° are negative ($\Delta H^\circ = -109 \text{ kJ mol}^{-1}$; $\Delta S^\circ = -155 \text{ J mol}^{-1}\text{K}^{-1}$)

ΔH° is negative because sigma bonds are being formed while pi bonds are being broken. Sigma bonds are of lower energy so reaction is exothermic

and

ΔS° is negative because there is a large change in number of molecules between the reactants and the products therefore there is a significant change in entropy and thus ΔS° value. Another contribution is because ethylene is a gas whereas the polymer is a solid. There is thus a reduction in degrees of freedom.

b) Calculate ΔG°
($\Delta H^\circ = -109 \text{ kJ mol}^{-1}$; $\Delta S^\circ = -155 \text{ J mol}^{-1}\text{K}^{-1}$)

$$\Delta G^\circ = \Delta H^\circ - T \Delta S^\circ$$

$$= -109 \text{ kJ mol}^{-1} - (298 \text{ K}) (-155 \text{ J mol}^{-1}\text{K}^{-1})$$

$$= -62.81 \text{ kJ mol}^{-1}$$

c) Calculate at which temperature polymerisation stops being spontaneous

$$\Delta G^\circ = \Delta H^\circ - T \Delta S^\circ$$

$$0 = \Delta H^\circ - T \Delta S^\circ$$

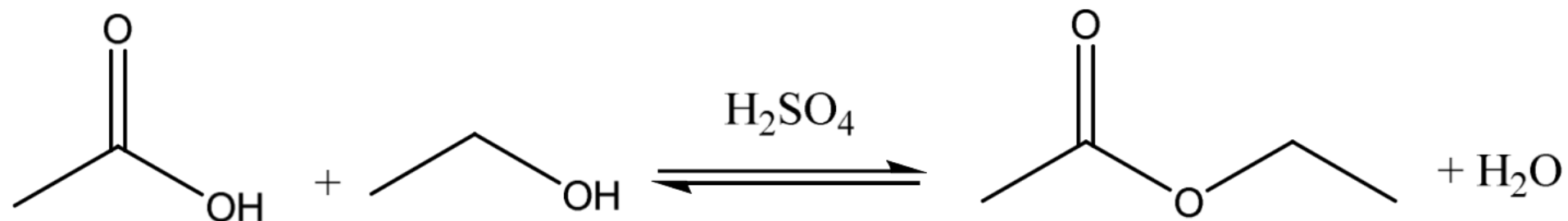
$$T_{\text{eq}} = (\Delta H^\circ) / (\Delta S^\circ)$$

$$T = 703 \text{ K}$$

d) What limitations does the calculation in c) have?

It will only be an "approximate temperature" because of the approximations we make about ΔH and ΔS not changing with temperature.

3) Consider the following reaction:



- a) Knowing $\Delta H^\circ = -2 \text{ kJ mol}^{-1}$ and $\Delta S^\circ = +8 \text{ J mol}^{-1}\text{K}^{-1}$, calculate the equilibrium constant, K , at 298 K.
- b) Which reaction will be spontaneous at conversions 0.3, 0.6 and 0.8
- c) Will your answer be the same if the starting point is twice as much ethanol (CCO) as acetic acid (CC(=O)O)?

a) Knowing $\Delta H^\circ = -2 \text{ kJ mol}^{-1}$ and $\Delta S^\circ = +8 \text{ J mol}^{-1}\text{K}^{-1}$, calculate the equilibrium constant, K , at 298 K.

$$\Delta G^\circ = \Delta H^\circ - T \Delta S^\circ = -RT \ln K$$

$$K = e^{\frac{-\Delta H^\circ + T \Delta S^\circ}{RT}} = e^{\frac{2 \text{ kJ mol}^{-1} + 298 \text{ K} (8 \text{ J mol}^{-1}\text{K}^{-1})}{(8.31 \text{ J mol}^{-1} \text{K}^{-1})298\text{K}}}$$

$$K=5.87$$

b) Which reaction will be spontaneous at conversions 0.3, 0.6 and 0.8

$$Q = \frac{x^2}{(1-x)^2} = \frac{0.3^2}{(0.7)^2} = 0.184$$

The Forward reaction because $Q < K$

$$Q = \frac{x^2}{(1-x)^2} = \frac{0.6^2}{(0.4)^2} = 2.25$$

The forward reaction because $Q < K$

$$Q = \frac{x^2}{(1-x)^2} = \frac{0.8^2}{(0.2)^2} = 16$$

The backward reaction because $Q > K$

Will your answer be the same if the starting point is twice as much

ethanol (CCO) as acetic acid (CC(=O)O)?

$$Q = \frac{x^2}{(2-x)(1-x)} = \frac{0.3^2}{(1.7)(0.7)} = 0.076$$

The Forward reaction because $Q < K$

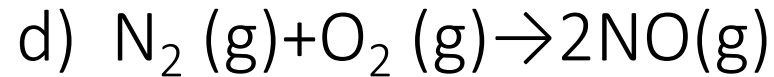
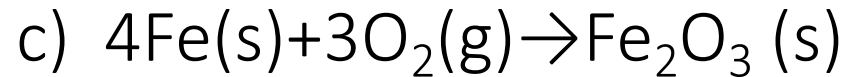
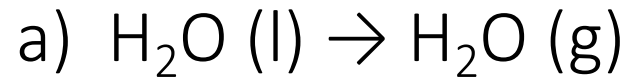
$$Q = \frac{x^2}{(2-x)(1-x)} = \frac{0.6^2}{(1.4)(0.4)} = 0.64$$

The forward reaction because $Q < K$

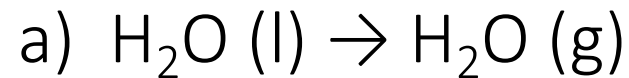
$$Q = \frac{x^2}{(2-x)(1-x)} = \frac{0.8^2}{(1.2)(0.2)} = 2.67$$

The forward reaction because $Q < K$

4) Predict whether the following reactions will occur with negative, positive or very small (close to 0) entropy change. Explain why:



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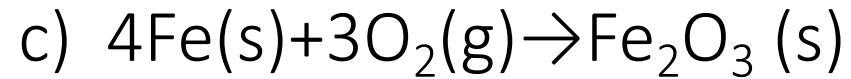
$\Delta S > 0$, Large : There is a change of phase from the liquid phase to the gas phase

4) Predict whether the following reactions will occur with negative, positive or very small (close to 0) entropy change. Explain why:



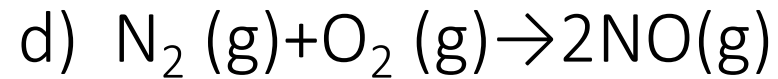
$\Delta S < 0$: There is a reduction in degrees of freedom from a dissolved state to a order solid

4) Predict whether the following reactions will occur with negative, positive or very small (close to 0) entropy change. Explain why:



$\Delta S < 0$: There is a reduction in degrees of freedom from gaseous O_2 molecules to a solid

4) Predict whether the following reactions will occur with negative, positive or very small (close to 0) entropy change. Explain why:



$\Delta S \approx 0$: We go from 2 gas molecules to 2 gas molecules.

5) Consider the reaction: $\text{CH}_4 (\text{g}) + 2\text{O}_2 (\text{g}) \rightarrow \text{CO}_2 (\text{g}) + 2\text{H}_2\text{O} (\text{g})$,
 $\Delta H^\circ = -802.3 \text{ kJ mol}^{-1}$; $\Delta S^\circ = -5 \text{ J mol}^{-1} \text{ K}^{-1}$

- a) Will this reaction be spontaneous at standard conditions?
- b) Calculate the equilibrium constant, K .
- c) What temperature will this reaction cease to be spontaneous?
- d) Discuss the validity of these calculations.

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 $\Delta H^\circ = -802.3 \text{ kJ mol}^{-1}$; $\Delta S^\circ = -5 \text{ J mol}^{-1} \text{ K}^{-1}$

a) Will this reaction be spontaneous at standard conditions?

$$\Delta G^\circ = \Delta H^\circ - T \Delta S^\circ = -802.3 \text{ kJ mol}^{-1} - (298 \text{ K})(-5 \text{ J mol}^{-1} \text{ K}^{-1})$$

$$\Delta G^\circ = -800.8 \text{ kJ mol}^{-1}$$

5) Consider the reaction: $\text{CH}_4 (\text{g}) + 2\text{O}_2 (\text{g}) \rightarrow \text{CO}_2 (\text{g}) + 2\text{H}_2\text{O} (\text{g})$,
 $\Delta H^\circ = -802.3 \text{ kJ mol}^{-1}$; $\Delta S^\circ = -5 \text{ J mol}^{-1} \text{ K}^{-1}$

b) Calculate the equilibrium constant, K .

$$\Delta G^\circ = \Delta H^\circ - T \Delta S^\circ = -RT \ln K$$

$$K = e^{\frac{-\Delta H^\circ + T \Delta S^\circ}{RT}}$$

$$K = 2.77 \times 10^{140}$$

5) Consider the reaction: $\text{CH}_4 (\text{g}) + 2\text{O}_2 (\text{g}) \rightarrow \text{CO}_2 (\text{g}) + 2\text{H}_2\text{O} (\text{g})$,
 $\Delta H^\circ = -802.3 \text{ kJ mol}^{-1}$; $\Delta S^\circ = -5 \text{ J mol}^{-1} \text{ K}^{-1}$

c) What temperature will this reaction cease to be spontaneous?

$$\Delta G^\circ = \Delta H^\circ - T \Delta S^\circ$$

$$0 = \Delta H^\circ - T \Delta S^\circ$$

$$T_{\text{eq}} = (\Delta H^\circ) / (\Delta S^\circ)$$

$$T = 160\,460 \text{ K}$$

5) Consider the reaction: $\text{CH}_4 (\text{g}) + 2\text{O}_2 (\text{g}) \rightarrow \text{CO}_2 (\text{g}) + 2\text{H}_2\text{O} (\text{g})$,
 $\Delta H^\circ = -802.3 \text{ kJ mol}^{-1}$; $\Delta S^\circ = -5 \text{ J mol}^{-1} \text{ K}^{-1}$

d) Discuss the validity of these calculations.

It will only be an "approximate temperature" because of the approximations we make about ΔH and ΔS not changing with temperature.